

## READING LIST: SPAN 581

Title: Research Methods: Open Science and Reproducible Research in Linguistics

01:940:581:01 - Spring 2019

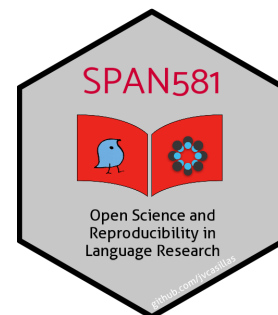
Meetings: AB 5190, 08:10–11:10

Professor: Joseph V. Casillas, PhD

Email: joseph.casillas@rutgers.edu

Office: AB 5174

Office hours: by appointment



**Note:** Readings must be completed *before* the class that occurs in the assigned week (i.e., readings assigned for week 2 will be discussed *in* the week 2 class, thus they should be read *before*.).

## Loading required namespace: bibtex

## Week 2:

### Reading

- Gonzales, K. and A. J. Lotto (2013). “A bafri, un pafri: Bilinguals’ pseudoword identifications support language-specific phonetic systems’”. In: *Psychological science* 24.11, pp. 2135–2142.
- Bohannon, J. (2015). “Many psychology papers fail replication test’”. In: *Science* 349.6251, pp. 910–911. ISSN: 0036-8075. DOI: 10.1126/science.349.6251.910. eprint: <http://science.sciencemag.org/content/349/6251/910.full.pdf>. <http://science.sciencemag.org/content/349/6251/910>.
- Open Science Collaboration (2015). “Estimating the reproducibility of psychological science’”. In: *Science* 349.6251. ISSN: 0036-8075. DOI: 10.1126/science.aac4716. eprint: <http://science.sciencemag.org/content/349/6251/aac4716.full.pdf>. <http://science.sciencemag.org/content/349/6251/aac4716>.
- Crüwell, S., J. van Doorn, A. Etz, et al. (2018). *8 Easy Steps to Open Science: An Annotated Reading List*. DOI: 10.31234/osf.io/cfzyx. [psyarxiv.com/cfzyx](https://psyarxiv.com/cfzyx).

### Additional resources

- Guides to better science: Data management
- GitHub for the user
- Good enough practices in scientific computing
- Using Git and GitHub
- The replication crisis (audio)

## Week 3:

### Reading

- Gonzales, K., K. Byers-Heinlein, and A. Lotto (2019). “How bilinguals perceive speech depends on which language they think they’re hearing’”. In: *Cognition* 182, pp. 318–330.
- van’t Veer, A. E. and R. Giner-Sorolla (2016). “Pre-registration in social psychology—A discussion and suggested template’”. In: *Journal of Experimental Social Psychology* 67, pp. 2–12.
- Wickham, H. and G. Grolemund (2016). *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. O’Reilly Media.
  - <https://r4ds.had.co.nz/r-markdown.html>
  - <https://r4ds.had.co.nz/r-markdown-formats.html>

- RMarkdown cheatsheet
- RMarkdown reference
- Getting started with RMarkdown
- Data science and R: How do I start?

## Week 4:

### Reading

- Casillas, J. V. and M. Simonet (2018). “Perceptual categorization and bilingual language modes: Assessing the double phonemic boundary in early and late bilinguals’”. In: *Journal of Phonetics* 71, pp. 51–64.
- Wickham, H. and G. Grolemund (2016). *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. O’Reilly Media.
  - <https://r4ds.had.co.nz/workflow-projects.html>
- Guides to better science: Reproducible code

### Additional resources

- What I wish I knew when I started R
- Getting help
- Field Guide to the R Ecosystem (highly recommended)

## Week 5:

### Reading

- Wickham, H. and G. Grolemund (2016). *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. O’Reilly Media.
  - <https://r4ds.had.co.nz/data-visualisation.html>
  - <https://r4ds.had.co.nz/workflow-basics.html>
- R for excel users

### Additional resources

- The tidyverse style guide
- Data structure basics
- ggplot2 cheatsheet
- Secrets to a happy graphing life
- A Compendium of Clean Graphs in R

## Week 6:



### Reading

- Wickham, H. and G. Grolemund (2016). *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. O'Reilly Media.
  - <https://r4ds.had.co.nz/transform.html>
  - <https://r4ds.had.co.nz/workflow-scripts.html>
  - <https://r4ds.had.co.nz/tidy-data.html>
  - <https://r4ds.had.co.nz/relational-data.html>

### Additional resources

- tidyR tutorial
- Tidy data
- Non-tidy data
- Creating new variables with mutate and ifelse (advanced)
- Getting help: reprex 1
- Getting help: reprex 2

## Week 7:

### Reading

- Wickham, H. and G. Grolemund (2016). *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. O'Reilly Media.
  - <https://r4ds.had.co.nz/data-import.html>
- papaja documentation: Read documentation, install package and play with demo example.
- R Markdown with papaja (video pt. 1)
- R Markdown with papaja (video pt. 2)

### Additional resources

- papaja: Reproducible APA manuscripts with R Markdown (skim)
- More papaja

## Week 8:

### Reading

- TBA

### Additional resources

- R Markdown presentations
- xaringan tutorial 1
- xaringan tutorial 2

## Week 9:

### Reading

- Kording, K. P. and B. Mensh (2017). “Ten simple rules for structuring papers’’. In: *bioRxiv*. eprint: <https://www.biorxiv.org/content/early/2017/05/23/088278.full.pdf>. <https://www.biorxiv.org/content/early/2017/05/23/088278>.
- An introduction to introductions

### Additional resources

## Week 10:

### Reading

- Psychopy tutorial

### Additional resources

## Week 11:

### Optional reading

- Anwyl-Irvine, A. L., J. Massonnié, A. Flitton, et al. (2018). *Gorilla in our Midst: An online behavioral experiment builder*. DOI: 10.31234/osf.io/kf5nu. [psyarxiv.com/kf5nu](https://psyarxiv.com/kf5nu).

### Additional resources

## Week 12:

### Reading

- Roettger, T. B. and D. Baer-Henney (2018). *Towards a replication culture in phonetic research: Speech production research in the classroom*. DOI: 10.31234/osf.io/q9t7c. [psyarxiv.com/q9t7c](https://psyarxiv.com/q9t7c).

### Additional resources

## Week 13:

### Reading

- McGuire, G. (2010). “A brief primer on experimental designs for speech perception research’’. In: *Laboratory Report 77*.

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Additional resources



## Week 14:

### Reading

- Guides to better science: Promoting your research
- Unity of science: Why linguistics faces bigger challenges than the replication crisis

Additional resources